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Systematic Review of Proteomics in Age-Related Macular Degeneration and Pathway Analysis of Significant Protein Changes

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Topic: Proteomics in age-related macular degeneration (AMD) research.

Clinical Relevance: AMD is the leading cause of blindness in industrialized countries, with a poorly understood pathogenesis. Proteomics can identify significantly altered proteins in AMD patients, aiding in understanding the disease's pathophysiology and potentially improving diagnosis or treatment strategies.

Methods: A systematic review of proteomic studies in AMD was conducted. Proteins significantly altered in dry and wet AMD and those tested as biomarkers were presented according to sample type (aqueous humor, plasma, urine, vitreous, retinal pigment epithelium/choroid, and tear film) and type of assay (mass spectrometry or aptamers) used in the individual studies. Proteins that exhibited at least a 2× fold change (FC) were further analyzed through functional enrichment analysis and protein–protein interaction networks (STRING database).

Results: Twenty-two studies (case-control and cohorts) with a total of 6932 participants were included. The included studies showed significant heterogeneity, and most of them lacked sufficient power. Results suggested that various proteins and pathways are implicated in AMD, and there were differences when comparing results from the individual studies and unbiased results. Although many proteins differed significantly between AMD and control groups, most exhibited less than a 2-FC. Functional analysis of proteins with ≥|2|-FCs (identified by unbiased proteomics in multiple biofluids) highlighted lipid metabolism and protease regulation pathways as central to both dry and wet AMD. Complement and coagulation cascades, chaperones, and glycolysis pathways were significant in wet AMD, whereas matrix remodeling pathways were enriched mostly in dry AMD.

Conclusion: Combining proteomics from various studies could reveal new protein–protein interaction networks and associated functional pathways that may suggest novel potential therapeutic targets for AMD. However, there is a scarcity of data available for early AMD from ocular biofluids, and it should be the aim of future proteomics studies.

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Age-related macular degeneration (AMD), the leading cause of blindness in the developed countries, is a complex condition that affects central vision and is influenced by various factors. Age is the most significant risk factor, but genetic and environmental factors also play a role. A large genome-wide association study[1] identified 52 independent variants at 34 different genomic loci, explaining 46.7% of the variability in AMD risk. The risk variants with the largest difference between late AMD patients and healthy controls reside within the complement factor H (CFH) genomic region and the ARMS2/HTRA1 locus on chromosome 10q26. However, for most variants, the effect size was small, and the exact mechanism by which these genetic changes contribute to the development of AMD remains unknown.

Proteins are undoubtedly fundamental players in all life processes and in the development and progression of diseases. Alterations in protein regulation and function, as well as their associated networks, are most likely to mediate the genetic risk of complex diseases. To gain insights into disease mechanisms, we can examine the full array of proteins found in a biological system, a method known as proteomics.[2] There are 2 major techniques in unbiased proteomics. Mass spectrometry (MS) provides quantitative data but requires relatively substantial sample volumes and sample preprocessing, resulting in high technical variability and identification of a small subset of proteins from the whole proteome. The newer, aptamer-based methods have the benefit of low sample volume requirement; they have low variability and high sensitivity and specificity, but they are not quantitative, and it is

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challenging to compare results between different runs.[3,4] Both proteomics approaches have been used in AMD on various biofluids or tissues (urine, plasma, aqueous humor [AH], vitreous, drusen, and retinal pigment epithelium [RPE]).[5] These studies have aimed to identify potential disease biomarkers, predict disease progression or response to treatment, and can identify potential therapeutic targets.

Objectives

The aim of the present study is to provide a comprehensive summary of all relevant studies on unbiased proteomics in AMD, identify consistent findings, highlight discrepancies, and underline potential gaps in the current research or methodological issues that will help guide future research.

Methods

Eligibility Criteria

Types of Participants. We included studies with participants irrespective of their age, origin, males, and females, with the diagnosis of AMD of all stages in 1 or both eyes.

Types of Interventions. We included proteomics studies either with traditional or aptamer-based methods. We included studies of all types of samples: plasma, AH, urine, vitreous, RPE or choroid, and tear film. Both observational and interventional studies were considered as eligible. Applied filters were English language, human species, publication date after 2012, and published papers only.

Types of Outcomes. We extracted qualitative data of proteomics in AMD and also extracted quantitative results. We tried to identify differences in proteomics of AMD patients compared with controls separately for each type of sample and type of AMD. We also tried to identify proteins that were validated as potential biomarkers.

Information Sources. We searched the Cochrane Database of Systematic Reviews, Central Register of Controlled Trials, including EMBASE and the International Clinical Trials Registry Platform, MEDLINE (PubMed), ScienceDirect, the metaRegister of Controlled Trials, Latin American and Caribbean Literature on Health Sciences, and ClinicalTrials.gov. Regarding gray literature, we searched Agency for Healthcare Research and Quality and Open Grey (System for Information on Grey Literature in Europe). We searched for all types of papers in order not to miss any relevant article, but only observational studies on the topic of interest were retrieved. We also searched the relevant reference lists for any missing studies. All databases have been searched from 2012 until 2022, and the last search was performed on November 30, 2022.

Search Strategy. The search strategy can be found in an appendix (Appendix 1, available at www.ophthalmologyscience.org). The accuracy of the search was evaluated by the Evidence-Based Checklist for the Peer Review of Electronic Search Strategies.[6]

Selection Process. The included studies were selected by 2 independent reviewers (OS and VC). Mendeley Desktop 1.17.9 was employed for handling the results. The titles and abstracts of all relevant electronic search results were examined. Duplicates were removed. For studies that met the inclusion criteria, the full-text reports were obtained and examined. Full-text reports compatible with all inclusion standards were examined again. Any disagreements between the reviewers were resolved by discussion. There was not any case of multiple reports of the same study or papers of the same study with different follow-up periods.

Data Collection Process

Data were extracted on Excel (Microsoft Corp) data collection forms by 2 independent review authors (OS and VC). Any disagreements were resolved by discussion. A pilot test of the form was carried out first in a few studies. The appropriate software (Review Manager 5.3) was used for data management.

Data Items

The following data were extracted:

- Study ID
- Methods: design; type of assay; sample type; pooling;
- Participants: number of participants; AMD stage; age; sex
- Outcomes: total number of proteins; proteins significantly altered; proteins significantly altered with fold change (FC) $\geq|2|$; upregulated and downregulated proteins; validated proteins; statistical methods; FC and *P* value (or adjusted *P* value); area under the curve (AUC) and 95% confidence interval (CI); missing data and number of missing participants (if applicable).

Risk of Bias in Individual Studies

Risk of bias assessment in the included studies was based on the ROBINS-I tool for assessing risk of bias in noninterventional studies.[7] Two independent reviewers (OS and VC) evaluated the included studies, and any disagreements were resolved by discussion.

Effects Measures

Fold change along with its *P* value and AUC with its 95% CI were extracted for all proteins, when available.

Synthesis Methods

The individual studies showed important heterogeneity, and it was impossible to proceed to meta-analysis. Significant protein changes were presented narratively according to AMD type and sample type, and functional enrichment analysis was performed for significantly altered proteins that exhibited at least a 2× FC between AMD patients and controls. STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) was employed for pathway and protein–protein interaction network analyses.

Results

Study Selection

Nine electronic databases and the reference lists of relevant studies have been searched. The MEDLINE search returned 229 results in the first place. Of 120 studies retrieved, 20 studies remained after title and abstract evaluation, and 17 of these were included after full-text examination. Three papers were retrieved from the Cochrane database, 2 from EMBASE, and 1 from the International Clinical Trials Registry Platform. One of them was a duplicate, one other was removed after abstract evaluation, and the third was available only as an abstract. ScienceDirect search retrieved

18 studies; 1 of them was a duplicate, and the others were removed after title and abstract evaluation. The other databases did not return any relevant results. The reference list search returned 5 additional results. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart (Fig 1) shows the selection process.[8]

Twenty-two studies were included in the qualitative synthesis: 20 case-control studies and 2 cohorts.[9–31] The most important characteristics of the included studies can be found in an appendix (Appendix 2, available at www.ophthalmologyscience.org, Characteristics of Included Studies). Figure 2 presents the risk of bias assessment of the included studies based on the ROBINS-I tool for noninterventional studies.

Study Characteristics

In total, 6932 participants were included in the present review. However, this number is skewed due to a single study,[16] which included 5457 patients. The total number of AMD patients was 2214 or 2915, as the same study provided 2 different AMD sample sizes.[16] Healthy participants (controls) were 466. Early/intermediate AMD patients were 1313 or 2014,[16] and geographic atrophy (GA) patients were 192. Neovascular AMD (nvAMD) patients were 710 in total. The mean (standard deviation) age for AMD patients was 75.1 (7.4) years and for healthy controls 70.7 (6.5) years. For early/intermediate AMD, the mean age of participants was 72.9 (5.7) years; for GA, 76.3 (9.9); and for nvAMD, 73.9 (7.8) years. The proportion of men in the individual studies was 45.7% for AMD patients and 47.3% for controls. For early/intermediate AMD, 37% of the participants were men; for GA, 65.8%; while for nvAMD, the percentage of men was 47.3. Nineteen studies used liquid chromatography (LC)-mass spectrometry (MS) for protein identification, and 3 studies used aptamers. Seven studies used AH, 5 studies used plasma, 3 studies used vitreous humor, 3 studies used submacular tissue, 2 studies used tear film, and only 1 used urine. Ten studies were designed in Asia, 7 in Europe, 4 in United States, and 1 in Australia. In 6 studies, the samples were pooled.[9,11,18–21] In 4 studies, the Age-Related Eye Disease Study system was used for AMD classification,[11,12,16,21] 1 study used Clinical Diagnostic Criteria for AMD classification system,[16] 1 study used the International System of Classification,[10] and 1 study used the Minnesota Grading System (MGS),[27] whereas the classification system was not reported for the rest of the studies.

Regarding the total number of proteins detected, the median was 4782 proteins for aptamer-based assays (range: 4001−4979, mean: 4587) and 234 for LC-MS (range: 6−769, mean: 83). Studies reported a median of 19 proteins (range: 4−185, mean: 37) being differentially expressed between AMD and controls. However, only 1 study had enough power to use Bonferroni correction,[16] whereas most studies reported unadjusted *P* values or false discovery rate (FDR). The median of proteins validated as biomarkers was 6 proteins (range: 2−39, mean: 9) and selection for proteins to be sent for validation was biased.

Risk of Bias in Individual Studies

The included studies were all judged as low risk of bias. Figure 2 presents the risk of bias assessment of the included studies based on the ROBINS-I tool.

Results of the Individual Studies

Tables 1 to 4 summarize key protein changes and validated biomarkers from individual studies or those most significantly altered in AMD patients compared with controls. Results are grouped by AMD type and sample type.

Proteomics of Urine in AMD Patients

Sivagurunathan et al[9] analyzed urine proteomes from 23 controls and 61 AMD patients (early/intermediate, GA, and nvAMD), pooling samples from 10 patients per group. With a $\geq\pm1.5\times$ FC cutoff, they identified 383 differentially expressed proteins: 224 upregulated and 107 downregulated in early/intermediate AMD, 98 upregulated and 44 downregulated in nvAMD, and 64 upregulated and 22 downregulated in GA. Only APOA1, tissue inhibitor of metalloproteinases 1 (TIMP1), and SERPINA1 (3/383) were validated; APOA1 and TIMP1 increased in nvAMD and GA but not early/intermediate AMD, with poor correlation between MS and enzyme-linked immunosorbent assay (ELISA) results. Key altered pathways across all AMD stages vs. controls were complement and coagulation cascades, cell adhesion molecules, lysosome-related proteins, extracellular matrix-receptor interaction, and carbohydrate metabolism.

Proteomics of Plasma in AMD Patients

Relevant search includes 5 major studies. Xu et al[10] used 2-dimensional (2D) gel electrophoresis to examine the proteomics of plasma from 13 patients with nvAMD, 11 with dry AMD, and 6 healthy controls. Only 28 of 219 proteins had more than a 2-FC among the different groups. Network analysis of these proteins revealed that redox balance, lipid metabolism, and cytoskeletal organization are involved in dry AMD, whereas complement/immune responses, tissue repair/angiogenesis, and cellular metabolism and stress play additional roles in wet-to-dry differentiation.

Kim et al[11] in 2014 evaluated pooled plasma from 20 nvAMD patients and 20 healthy controls. Only 35 of 173 proteins that displayed greater than twofold differential expression between nvAMD and controls reached statistical significance by unadjusted G-test. Ingenuity pathway analysis performed by the authors identified cellular growth and proliferation (10.3%), cell death (9.4%), cellular movement (8.1%), and cell-to-cell signaling and interactions (7.2%) as the top pathways. Only 7 of 26 proteins tested for validation by ELISA or western blot showed significantly different concentrations between patients with AMD and controls. From those, vinculin was tested further in 43 nvAMD, 10 early AMD (eAMD), and 60 controls. The authors concluded vinculin could distinguish between controls and AMD but not between exudative and early AMD. Vinculin,

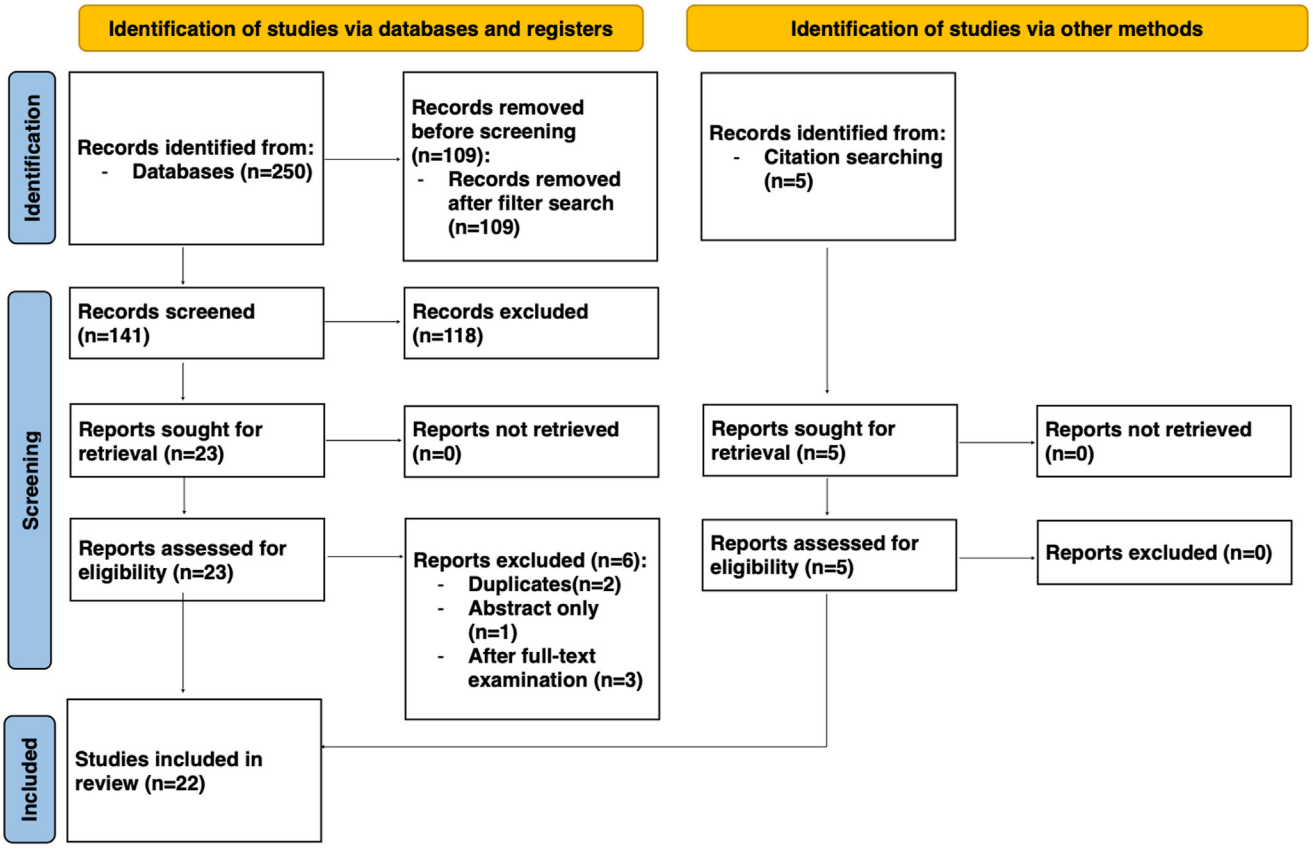


Figure 1. PRISMA 2020 flow diagram for new systematic reviews, which included searches of databases, registers, and other sources. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

a structural protein involved in cell adhesion, stability, and signal transduction, is also studied as a biomarker in cell migration and adhesion studies, particularly in cancer, cardiac function, and mechanotransduction.

In a subsequent study, the same group[12] validated phospholipid transfer protein and mannan-binding lectin-associated serine protease 1 using plasma from 58 nvAMD, 41 eAMD, and 61 controls in the first step, and plasma from 29 nvAMD, 8 eAMD, and 29 controls in the second step. They found that phospholipid transfer protein was upregulated in both early AMD and nvAMD, while mannan-binding lectin-associated serine protease 1 was significant in nvAMD. Phospholipid transfer protein binds phospholipids and facilitates their transfer between lipoproteins, including reverse cholesterol transfer. Elevated phospholipid transfer protein levels are associated with increased cardiovascular risk. Mannan-binding lectin-associated serine protease 1, a serine protease in the complement system's lectin pathway, is linked to autoimmune diseases and thrombosis. Phospholipid transfer protein showed a high AUC (0.923) in discriminating control vs. any AMD but was less successful in separating eAMD from nvAMD (AUC 0.711).

Aptamer-based protein identification, a recent proteomics method, involves the selection of short synthetic single-strand DNA or RNA molecules through an iterative process called SELEX.[13] Aptamer assays can measure significantly more proteins than MS (4000–7000 compared to <1000), yet they are still significantly less than all known proteins, estimated to be about 25 000.

In 2019, Lynch et al[14] used the SOMAscan assay to measure 4001 proteins in plasma from 10 nvAMD patients, 10 GA patients, and 10 controls. Unadjusted $P < 0.001$ revealed differential regulation of vinculin (upregulated in nvAMD) and CD177 (downregulated in nvAMD), as well as NRG4 (upregulated in GA) and sICAM-1 (downregulated in GA) compared with controls. Pathway analysis revealed differences between nvAMD and GA. The most important pathways of nvAMD include cargo trafficking to the periciliary membrane, FGFR3b ligand binding and activation, and VEGF-related pathways. In contrast, GA's most important pathways include ErbB signaling, cell death and survival, core cellular signaling processes, development, protein stability, and metabolic support. Complement factors were not differentially expressed in these advanced-stage AMD patients.

In 2020,[15] the same group used an updated aptamer library to evaluate 4979 proteins in 39 patients with intermediate AMD and subdrusenoid deposit (SDD) and 70 patients with intermediate AMD alone. They found 2 proteins (TCL1A and CNDP1) slightly lower (~1.3×) in

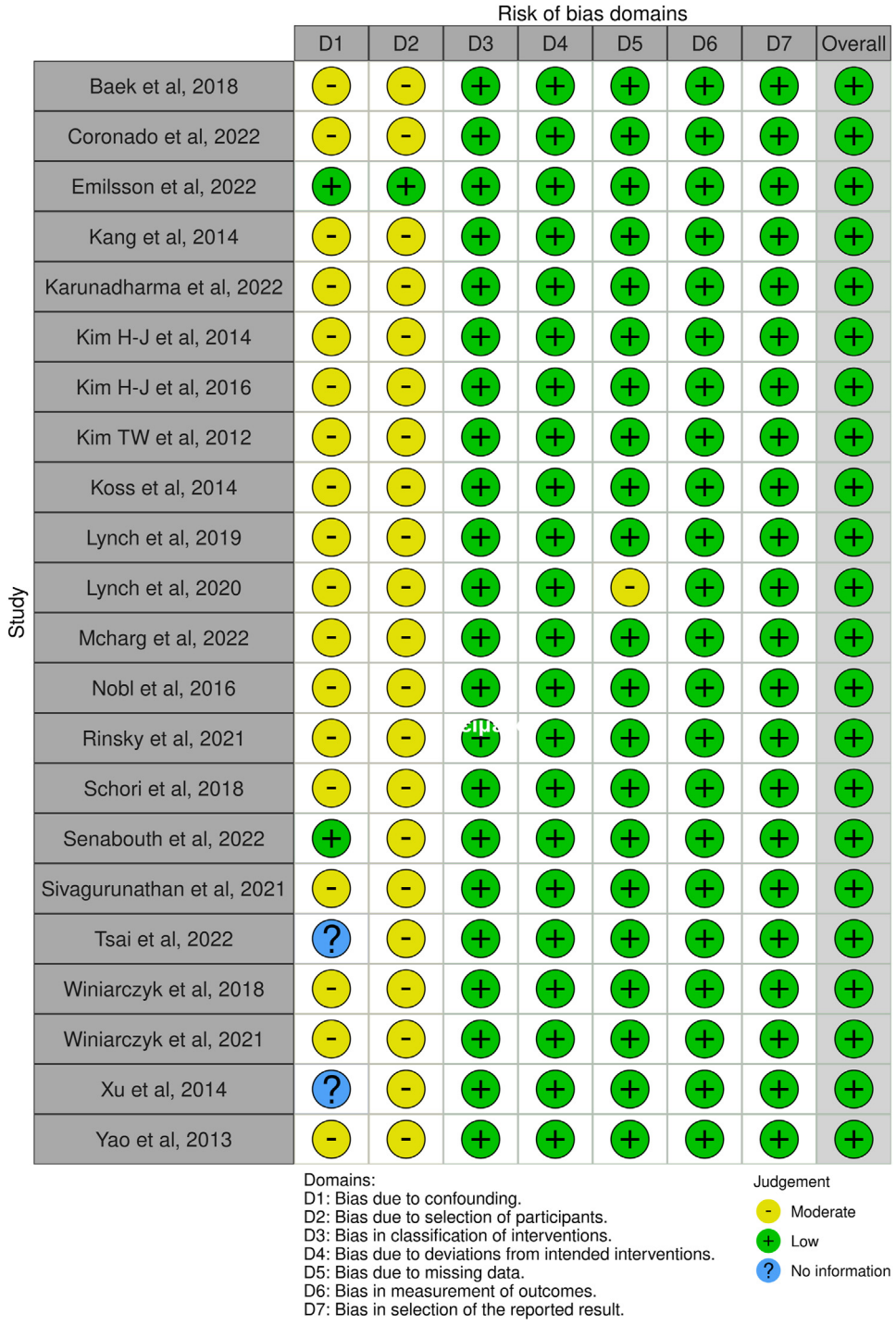


Figure 2. Risk of bias assessment in the individual studies.

patients with SDD, but neither was significant after multiple comparisons. The authors examined if clinical or proteomic information could predict progression to more advanced AMD stages (GA, nvAMD, or both). They found that 4 proteins (SAP3, lysozyme, RNAS6, and TFF3) had significant unadjusted association with the time to conversion (hazard ratio = 10.1, 5.6, 3.9, and 3.0 respectively). These associations were mainly driven by nvAMD. Pathway analysis for conversion suggested the top 4 pathways: tumor necrosis factor binding, digestion and absorption, activin, and transforming growth factor-β family signaling. However, none of these protein or pathway associations were significant after multiple comparison testing.

Table 1. Selected Candidate Proteins (AMD All Types)

Groups	Pts	Assay Type	Proteins	Fold Change	*P* Value	AUC	95% CI	Statistical Methods
Aqueous humor								
Rinsky et al, 2021								
• CTRL	N = 30	MS1, ELISA	Clusterin	1.06	0.032	-	-	>2 and <0.5 fold cut-off and
• AMD	N = 40		SerpinA4	0.98	0.62	-	-	*P* value <0.05
			TF	0.65	0.14	-	-	
Plasma								
Kim et al, 2016								
• CTRL	N = 90	LC-MS/MS,	MASP1	1.13	0.001	0.722	-	Regression analysis with
• AMD	N = 136	ELISA, WB	PLTP	11	0.001	0.923	-	Bonferroni correction (*P* value <0.05)
Emilsson et al, 2022								
• CTRL	N = 3430 (or 4131)	Aptamers,	NDUFS4	0.57	7.3E-14	0.696	-	Regression analysis,
• AMD	N = 2027 (or 1326)	MS, ELISA	CFHR1	1.56	2.5E-05	0.694	-	Bonferroni-adjusted
			FUT5	1.51	0.00016	0.691	-	*P* value, $P < 1.2*10^{-5}$
			TST	0.67	1.3E-07	0.691	-	
			BAMBI	0.66	0.00255	0.692	-	
Urine								
Sivagurunathan et al, 2021								
• CTRL	N = 23	MS, ELISA	SERPINA1	2	0.049	0.65	-	>1.5 and <0.6 fold cut-off
• AMD	N = 61		TIMP1	2.64	0.01	0.65	-	and *t* test or Mann
			APOA1	8.9	0.003	0.66	-	–Whitney test ($P < 0.05$)

AMD = age-related macular degeneration; AUC = area under the curve; CI = confidence interval; CTRL = controls; ELISA = enzyme-linked immunosorbent assay; LC-MS = liquid chromatography-mass spectrometry; MASP1 = mannan-binding lectin-associated serine protease 1; MS = mass spectrometry; PLTP = phospholipid transfer protein; Pts = patients; TIMP = tissue inhibitor of metalloproteinases; WB = western blot.

Emilsson et al[16] (2022) used the SOMAscan platform to measure >4000 serum protein levels in 5457 individuals from the prospective Age, Gene/Environment Susceptibility Reykjavik Study cohort. This is the only study that has the statistical power to detect significance by Bonferroni correction. Twenty-eight proteins were significantly associated with AMD, with varying associations depending on the classification scheme for early/intermediate AMD (Jonasson et al[32] vs. Holliday et al[33]). Six proteins were found differentially expressed in advanced AMD: NDUFS4, CFHR1, FUT5, TST, CFHR5, and BIRC2/cIAP1 if the Jonasson classification was used vs. 13 (NDUFS4, TST, DLL3, KREMEN2, CFHR1, TNFRSF14, CCL1, BAMBI, RGS8, ST6GALNAC1, DLL3, CRABP1, CXCL17, and BPIFB1) with the Holliday definition of the earlier stages of AMD.

Analysis of protein pathways showed the importance of mitochondrial function in both early and late AMD. Specifically, NDUFS4, a protein that assembles and stabilizes complex I in the mitochondrial electron transport chain, and TST, a mitochondrial matrix enzyme involved in cyanide detoxification, iron–sulfur protein formation, and modification of sulfur-containing enzymes, are highlighted as key players. Beyond mitochondrial processes, other significant pathways in early AMD include Notch and Wnt signaling. In these, DLL3 acts to inhibit Notch signaling, while KREMEN2 modulates Wnt signaling by blocking Dkk2's activation of LRP6, thereby enhancing inhibition of the Wnt/Frizzled pathway.

In advanced AMD, the leukocyte–endothelial adhesion pathway emerges as a critical mechanism, alongside mitochondrial pathways. FUT5, a protein involved in Lewis antigen synthesis, associates strongly with disease progression in these advanced stages.

When late AMD was analyzed as a combination of nvAMD and GA, CFHR1 and CFHR5 were significantly associated with late AMD, whereas C3, CFH, and CFB show no such connection. Further analysis identified PRMT3, an arginine methyltransferase that regulates ribosomal activity by methylating the 40S ribosomal subunit protein RPS2, as a potential predictor of progression to GA, with an AUC of 0.76 and a *P* value of 0.025.

Using a 2-sample Mendelian randomization approach, CFHR1, CFHR5, and FUT5 showed weak causal relationships (FDR <0.05). Elevated CFHR1 increased AMD risk, while elevated NDUFS4 levels conferred protection against late-stage AMD. A follow-up Mendelian randomization analysis incorporating 1327 aptamers with cis-acting genetic instruments identified 8 additional AMD-related proteins, including C3, CFI, allograft inflammatory factor 1, and VTN. Allograft inflammatory factor 1 stands out, significantly associated with AMD outcomes and maintaining directional consistency after multiple testing correction. The ARMS2/HTRA1 locus on chromosome 10, with its polymorphism rs3750846, was not investigated due to the SOMAscan platform's inability to detect ARMS2 or HTRA1 proteins.

Proteomics of AH in AMD Patients

Literature search found 8 studies on AH proteome analysis, all with small sample sizes and mostly comparing nvAMD with controls.

Kim et al[17] used 3 nvAMD and 3 controls for comprehensive proteome analysis and multiple reaction

Table 2. Selected Candidate Proteins (Dry AMD-Early/Intermediate)

Groups	Pts	Assay Type	Protein	Fold Change	P Value	AUC	95% CI	Statistical Methods
Aqueous humor								
Baek et al, 2018								
• CTRL	N = 2	SWATH-MS,	SERPINA4	2.67	0.003	-	-	ANOVA, adjusted
• e/iAMD (drusen)	N = 11	WB, ELISA	LUM	8.28	0.023	-	-	*P* value <0.05
			KERA	2.87	0.032	-	-	
			APOA1	1.76	0.041	-	-	
			CFHR2	0.44	0.027			
			PEDF	0.39	0.031			
			TIMP1	0.54	0.006			
Vitreous								
Schori et al, 2018								
• CTRL	N = 9	LC-MS/MS,	HEM2	5.5	0.035	-	-	Log2 (FC) >0.58 and
• Dry AMD	N = 6	ELISA	RNAS1	2.43	0.01	-	-	ANOVA with adjusted
			CPVL	2.13	0.05	-	-	*P* value <0.05
			CHLE	1.68	0.03	-	-	
			VGFR1	−1.24	0.72	-	-	
			SODC	1.72	0.15	-	-	
			VGFR2	−1.96	0.01	-	-	
			CFAI	−1.36	0.34	-	-	
RPE, choroid								
Mcharg et al, 2022								
• CTRL (high risk Chr1)	N = 8	iTRAQ Labeling,	ITIH3	3.572	0.0043	-	-	Bayesian mixed effects
• eAMD	N = 6	LC-MS	RNASE4	2.174	0.0017	-	-	model, gFDR <0.05
			PLA2	2.081	0.0015	-	-	
			SFRP5	2.054	0.0003	-	-	
Plasma								
Xu et al, 2014								
• CTRL	N = 6	MALDI-TOF/	Cofilin-1	3.23	-	-	-	*t* test with *P* value <0.05
• Dry AMD	N = 11	TOF MS	APOH β2-GPI	0.44	-	-	-	
			APOA4	0.23	-	-	-	
			SOD	0.43	-	-	-	
			APOA1	2.33	-	-	-	
			VDBP	0.16	-	-	-	
Emilsson et al, 2022								
• CTRL	N =	Aptamers, MS,	NDUFS4	0.616	1.4E-08	0.674	-	Regression analysis,
• eAMD	3430 (or 4131)	ELISA	BAMBI	1.51	0.00357	0.672	-	Bonferroni-adjusted
	N = 1755		ST6GALNAC1	1.46	0.00456	0.671	-	*P* value, $P < 1.2*10^{-5}$
	(or 1054)		FUT5	1.44	0.00747	0.67	-	
			CEBPB	1.48	0.01239	0.669	-	
Urine								
Sivagurunathan et al, 2021								
• CTRL	N = 23	MS, ELISA	SERPINA1	2	0.044	-	-	>1.5 and <0.6 fold cut-
• eAMD	N = 28		TIMP1	2.18	0.12	-	-	off and *t* test or Mann
			APOA1	4.02	0.1	-	-	−Whitney test
								($P < 0.05$)

AMD = age-related macular degeneration; ANOVA = analysis of variance; AUC = area under the curve; CI = confidence interval; CTRL = controls; e/iAMD = early/intermediate age-related macular degeneration; eAMD = early age-related macular degeneration; ELISA = enzyme-linked immunosorbent assay; FC = fold change; FDR = false discovery rate; gFDR = global False Discovery Rate; LC-MS = liquid chromatography-mass spectrometry; MS = mass spectrometry; Pts = patients; RPE = retinal pigment epithelium; TIMP = tissue inhibitor of metalloproteinases; WB = western blot.

Table 3. Selected Candidate Proteins (GA)

Groups	Pts	Assay Type	Protein	Fold Change	*P* Value	AUC	95% CI	Statistical Methods
Aqueous humor								
Rinsky et al, 2021								
• CTRL	N = 30	MS1, ELISA	Clusterin	1.11	0.0045	0.73	-	>2 and <0.5 fold cut-off and *P* value <0.05
• GA	N = 15		SerpinA4	1.09	0.04	0.61	-	
			TF	0.79	0.52	-	-	
RPE, choroid								
Senabouth et al, 2022								
• CTRL	N = 36	LC-ESI-MS/MS,	TIMP3	1.33	0.0001	-	-	*t* test *P* value <0.05 and FC ≥1.1
• GA	N = 43	iPSC-derived	ES8L2	1.28	0.0001	-	-	
		RPE cells	C1R	1.32	0.001	-	-	
			STMN2	0.75	0.001	-	-	
Plasma								
Lynch et al, 2019								
• CTRL	N = 10	Aptamers	NRG4	1.19	<0.01	-	-	Bonferroni adjustment ($P < 0.001$)
• GA	N = 10		sICAM-1	0.53	<0.01	-	-	
Urine								
Sivagurunathan et al, 2021								
• CTRL	N = 23	MS, ELISA	SERPINA1	-	0.87	-	-	>1.5 and <0.6 fold cut-off and *t* test or Mann–Whitney test ($P < 0.05$)
• GA	N = 12		TIMP1	4.51	0.02	-	-	
			APOA1	11.81	0.008	-	-	

AUC = area under the curve; CI = confidence interval; CTRL = controls; ELISA = enzyme-linked immunosorbent assay; FC = fold change; GA = geographic atrophy; iPSC = induced pluripotent stem cell; LC-MS = liquid chromatography-mass spectrometry; MS = mass spectrometry; Pts = patients; RPE = retinal pigment epithelium; TIMP = tissue inhibitor of metalloproteinases.

monitoring MS. They then performed a systematic multiple reaction monitoring MS assay on 7 highly abundant differentially expressed proteins from these groups in an independent set of 11 AH samples (6 from AMD and 5 from cataract controls). A total of 132 proteins were identified in AMD and 102 in the control group (80 common) during the comprehensive proteomics phase. Complement and antioxidant pathway proteins were slightly enriched in nvAMD, but statistics were not possible due to the small sample sizes. Seven proteins were tested for validation and discriminatory ability between nvAMD and controls. TGFBI, clusterin, and ceruloplasmin had the highest discriminatory values (AUC: 0.799, 0.781, and 0.778, respectively).

Yao et al[18] examined AH from 6 patients with nvAMD and 6 controls by 2D electrophoresis. They identified 78 protein spots with 68 differentially regulated proteins. The pathways involved include lipid metabolism and transport (LCN1 and AZGP1) as well as immune responses (CCL24, PDCD1LG2, and COX-2). Only LCN1 (a lipid-binding protein) and CRYAA (a chaperone) were validated by ELISA. LCN1 was significantly higher in nvAMD, while CRYAA was undetectable in both groups.

Kang et al[19] analyzed proteins from exosomes of 9 patients with nvAMD and 9 controls in AH, pooling the AH data due to the small number of exosomes per sample. They performed LC-multiple reaction monitoring for 6 proteins (actin, myosin-9, HSP70, cathepsin D, cytokeratin 8, and cytokeratin 14) in AH from 14 nvAMD patients and 6 controls. All 6 proteins were elevated by about 1.5-fold. Cytokeratin 8 had the highest AUC (0.929, CI 0.786–0.988) in distinguishing nvAMD from controls.

Baek et al[20] used SWATH-MS to detect AH proteins from 6 patients (2 controls, 2 classic drusen, and 2 subretinal drusenoid deposits). They identified 41 differentially regulated proteins with a 2-FC over control and a q-value <0.05. Four proteins (LUM, KERA, DSP, and SERPINA4) were upregulated and 3 proteins (VCAN, DKK3, and SPP1) were downregulated in both classic drusen and SDD. Three proteins were upregulated in SDD but downregulated in classic drusen. Six were downregulated in classic drusen but not in SDDs and 13 downregulated in SDD but not in classic. Finally, 8 proteins were upregulated in SDD but not in classic drusen.

Rinsky et al[21] analyzed the proteome of 20 patients (10 nvAMD and 10 controls) using MS1 with a 2-FC threshold. They identified 239 upregulated and 86 downregulated proteins. Due to pooling, no statistics were performed. Database for annotation, visualization, and integrated discovery functional analysis identified 3 functional clusters of proteins significantly upregulated in nvAMD: platelet degranulation, cellular protein metabolic processes, and complement and coagulation cascades. In an ELISA validation of 9 proteins (clusterin, serpin A4, IGFBP-2, IL1 R3, RBP4, CFD, CRP, FETUB, and TF) using a new set of 20 controls (15 atrophic AMD and 15 nvAMD), none of the proteins reached a 2-fold difference, and most were close to a 1.1-fold difference.

Coronado et al[22] conducted a pilot study with 5 nvAMD patients who responded well to anti-VEGF treatment, 5 who

Table 4. Selected Candidate Proteins (nvAMD)

Groups	Pts	Assay Type	Protein	Fold Change	P Value	AUC	95% CI	Statistical Methods
Aqueous humor								
Kim et al, 2012								
• CTRL	N = 8	LC-ESI-MS/MS, WB	TGFB1	-	0.0096	0.799	(0.596, 0.929)	*t* test with *P* value <0.05
• nvAMD	N = 9		Ceruloplasmin	2.6	0.0058	0.778	(0.573, 0.916)	
			PEDF	2.5	0.0312	0.667	(0.456, 0.838)	
			SERPING1	2.2	0.045	0.656	(0.446, 0.830)	
			Clusterin	-	-	0.781	(0.577, 0.918)	
			Cathepsin D	-	-	0.583	(0.375, 0.722)	
Yao et al, 2013								
• CTRL	N = 6	MALDI-TOF	LCN1	57.09	0.01	-	-	*t* test with *P* value < 0.05
• nvAMD	N = 6	MS/MS, ELISA						
Kang et al, 2014								
• CTRL	N = 18	LC-ESI-MS/MS, WB	Actin	3.24	0.0005	0.875	(0.716, 0.963)	*t* test with *P* value <0.05
• nvAMD	N = 26		Myosin-9	1.73	0.0152	0.756	(0.579, 0.886)	
			HSP70	1.41	0.0463	0.685	(0.503, 0.832)	
			Cathepsin D	1.62	0.0363	0.702	(0.522, 0.846)	
			Cytokeratin 8	2.09	0.0002	0.929	(0.786, 0.988)	
			Cytokeratin 14	2.10	0.0084	0.774	(0.598, 0.899)	
Rinsky et al, 2021								
• CTRL	N = 30	MS1, ELISA	Clusterin	0.99	0.87	-	-	>2 and <0.5 fold cut-off
• nvAMD	N = 25		SerpinA4	0.93	0.05	0.68	-	and *P* value <0.05
			TF	0.47	0.13	-	-	
Tsai et al, 2022								
• CTRL	N = 25	LC-MS/MS	APOA1	2.32	1.81E-05	-	-	PCA, *t* test with *P* value
• nvAMD	N = 28		Clusterin	2.60	8.10E-03	-	-	<0.05
			Complement C3	3.16	2.84E-02	-	-	
			Opticin	3.93	4.40E-03	-	-	
			PTGS2	0.50	4.86E-02	-	-	
Vitreous								
Koss et al, 2014								
• CTRL	N = 19	LC-MS/MS, WB	A1 antitrypsin	2.35	0.02	-	-	Benjamini–Hochberg
• nvAMD	N = 77		APOA1	1.32	0.27	-	-	adjusted *P* value <0.05,
			RBP3	1.10	1.10	-	-	Mann–Whitney or
			Serotransferrin	1.16	1.16	-	-	Wilcoxon test
			Transthyretin	1.57	1.57	-	-	
Nobl et al, 2016								
• CTRL	N = 24	LC-MS/MS, ELISA	Clusterin	2.19	3.14E-02	0.747	(0.603, 0.890)	Benjamini–Hochberg,
• nvAMD	N = 108		Opticin	0.37	1.49E-01	0.656	(0.478, 0.835)	adjusted *P* value
			PEDF	1.98	3.14E-02	0.514	(0.341, 0.687)	<5.00E-02, Kruskal
			PH2D	1.5	1.49E-01	0.712	(0.519, 0.904)	–Wallis, Mann
								–Whitney, and *t* test
Schori et al, 2018								

(*Continued*)

Table 4. (Continued.)

Groups	Pts	Assay Type	Protein	Fold Change	P Value	AUC	95% CI	Statistical Methods
• CTRL (ERM) • nvAMD	N = 9 N = 10	LC-MS/MS, ELISA	RNAS1 CPVL CHLE VGFR1 SODC VGFR2 CFAI	1.69 1.91 1.11 8.23 1.57 109.32 1.19	0.01 0.01 0.80 0.05 0.03 0.14 0.41	- - - - - - -	- - - - - - -	Log2 (FC) >0.58 and ANOVA with unadjusted *P* value <0.05
Plasma								
Xu et al, 2014								
• CTRL • nvAMD	N = 6 N = 13	MALDI-TOF/TOF MS	HLA-DRB1-5 Plasma RBP APOH (β2-GPI) HSPB1 UBC(E2) enzyme GCAP SOD PRDX3 isoform β	5.65 9.09 2.17 0.1 0.25 0.17 0.009 0.04	- - - - - - - -	- - - - - - - -	- - - - - - - -	*t* test with *P* value <0.05
Kim et al, 2014								
• CTRL • nvAMD	N = 100 N = 101	LC-MS/MS, ELISA, WB	Vinculin S100A9 S100A8 CXCL7 TNX TPI ZPI	9.07 2.15 19.15 2.15 0.08 9.07 4.23	<0.1 <0.05 <0.05 <0.05 <0.05 <0.1 <0.05	0.871 - - - - - -	- - - - - - -	G value >3.81 ($P < 0.05$) or G value >2.71 ($P < 0.1$)
Lynch et al, 2019								
• CTRL • nvAMD	N = 10 N = 10	Aptamers	Vinculin CD177	1.98 0.35	0.001 0.001	- -	- -	Bonferroni adjustment ($P < 0.001$)
Urine								
Sivagurunathan et al, 2021								
• CTRL • nvAMD	N = 23 N = 21	MS, ELISA	SERPINA1 TIMP1 APOA1	2 2.82 15.59	0.027 0.01 0.003	- - -	- - -	>1.5 and <0.6 fold cut-off and *t* test or Mann–Whitney test ($P < 0.05$)
Tear film								
Winiarczyk et al, 2021								
• CTRL • nvAMD	N = 15 N = 15	MALDI-TOF MS	RALDH 1 ATP dependent translocase ABCB1 Alpha-enolase	2.072 2.193 1.476	0.011 0.025 0.022	- - -	- - -	*t* test with *P* value <0.05 and Rts >1.5 or <0.6

ANOVA = analysis of variance; ATP = adenosine triphosphate; AUC = area under the curve; CI = confidence interval; CTRL = controls; ELISA = enzyme-linked immunosorbent assay; ERM = epiretinal membrane; FC = fold change; LC = liquid chromatography; MS = mass spectrometry; nvAMD = neovascular age-related macular degeneration; PCA = principal component analysis; Pts = patients; Rts = ratios; TIMP = tissue inhibitor of metalloproteinases; WB = western blot.

Table 5. Biological Process (Gene Ontology), Unbiased Proteomics

AMD Type	GO Term	Description	Strength	Signal	FDR
Dry (e/i) AMD	GO:0010466	Negative regulation of peptidase activity	1.19	1.62	6.10E-08
	GO:0010951	Negative regulation of endopeptidase activity	1.17	1.54	1.44E-07
	GO:0045861	Negative regulation of proteolysis	1.08	1.47	8.48E-08
	GO:0051346	Negative regulation of hydrolase activity	1.07	1.44	9.85E-08
	GO:0052547	Regulation of peptidase activity	1.0	1.34	1.17E-07
nvAMD	GO:0010951	Negative regulation of endopeptidase activity	1.21	2.17	1.73E-11
	GO:0007599	Hemostasis	1.25	2.07	3.10E-10
	GO:0051346	Negative regulation of hydrolase activity	1.08	1.95	1.73E-11
	GO:0007596	Blood coagulation	1.23	1.91	2.14E-09
	GO:0052548	Regulation of endopeptidase activity	1.02	1.74	1.20E-10

AMD = age-related macular degeneration; e/i = early/intermediate; FDR = false discovery rate; GO = gene ontology; nvAMD = neovascular age-related macular degeneration.

did not, and 5 controls. Overall, 185 of 2366 proteins were statistically significant (FDR <1%) and allowed for distinguishing the 3 groups by hierarchical clustering analysis. They performed partial least squares discriminant analysis to obtain variable importance projection scores and identify 30 most discriminant proteins. The top 10 were catalase, beta crystallin B1, heat shock beta 1, sirtuin, alpha crystallin A2, beta crystallin A3, carbonyl reductase, alpha crystallin B, crystallin gamma D, and beta crystallin A4. The majority of differentially expressed proteins converge on 3 main biological process pathways: homeostasis and coagulation effectors, lipid metabolism effectors, and angiogenesis pathway effectors.

Tsai et al[23] compared the exosomal protein profile of 28 nvAMD patients to 25 controls using principal component analysis. They found partial overlap between AMD and control groups, with high individual variation in AMD. Gene ontology analysis revealed similar biological processes, but the lipoprotein metabolic process was enriched in AMD. Comparison of exosomal proteins between nvAMD and controls revealed 10 significantly different peptides (upregulated: APOA1, OPTC, CLU, C3; downregulated: PTGDS, Q6MZU6). They also analyzed the exosome profile of 2 nvAMD patients before and after anti-VEGF injection, finding decreased amounts of SERPINA1 and AZGP1.

Proteomics of Vitreous in AMD Patients

There were only 3 studies that examined the vitreous of AMD patients. There were none with GA patients.

Koss et al[24] analyzed the vitreous of 73 advanced untreated nvAMD and 15 controls using capillary electrophoresis-MS and LC-MS. They identified 97 unique proteins, with 19 differentially upregulated in nvAMD and 12 after multiple testing correction. Five proteins (alpha-1-antitrypsin, apolipoprotein A-1, retinol-binding protein 3, serotransferrin, and transthyretin) were randomly selected from the set of 19 for validation by western blot using 4 new patients per group. Only alpha-1-antitrypsin reached statistical significance, while transthyretin and apolipoprotein A-1 showed a nonsignificant increase in AMD.

Nobl et al[25] performed a capillary electrophoresis-MS proteomics biomarker discovery using 72 nvAMD patients and 16 controls, followed by validation through capillary electrophoresis-MS proteomics in 36 nvAMD and 8 controls and ELISA testing in 8 nvAMD and 6 controls. Clusterin and pigment epithelium-derived factor were upregulated after multiple testing and are thought to be involved in nvAMD. The receiver operating characteristic analysis for clusterin showed an AUC of 0.747 (CI of 0.603−0.890) for predicting nvAMD.

Schori et al[26] analyzed the vitreous of 16 patients (6 dry AMD and 10 nvAMD) against 9 epiretinal membrane

Table 6. Molecular Function (Gene Ontology), Unbiased Proteomics

AMD Type	Pathway	Description	Strength	Signal	FDR
Dry (e/i) AMD	GO:0004866	Endopeptidase inhibitor activity	1.26	1.66	1.22E-07
	GO:0061134	Peptidase regulator activity	1.19	1.58	1.22E-07
	GO:0004867	Serine-type endopeptidase inhibitor activity	1.32	1.16	6.44E-05
nvAMD	GO:0004866	Endopeptidase inhibitor activity	1.25	2.03	5.44E-10
	GO:0004857	Enzyme inhibitor activity	1.01	1.67	5.44E-10
	GO:0016209	Antioxidant activity	1.32	1.15	4.86E-05
	GO:0002020	Protease binding	1.13	1.0	9.90E-05
	GO:0004867	Serine-type endopeptidase inhibitor activity	1.21	0.99	0.00016

AMD = age-related macular degeneration; e/i = early/intermediate; FDR = false discovery rate; GO = gene ontology; nvAMD = neovascular age-related macular degeneration.

Table 7. Kyoto Encyclopedia of Genes and Genomes Pathways, Unbiased Proteomics

AMD Type	Pathway	Description	Strength	Signal	FDR
Dry (e/i) AMD	hsa:04610	Complement and coagulation cascades	1.33	1.06	0.00018
	hsa:04514	Cell adhesion molecules	1.03	0.48	0.0209
	hsa:04975	Fat digestion and absorption	1.33	0.41	0.0499
nvAMD	hsa04610	Complement and coagulation cascades	1.44	2.13	2.87E-09
	hsa00010	Glycolysis/gluconeogenesis	1.15	0.41	0.0385

AMD = age-related macular degeneration; e/i = early/intermediate; FDR = false discovery rate; nvAMD = neovascular age-related macular degeneration.

controls. Performing gene ontology (GO) biological process analysis, they found stress response as the strongest enrichment set for dry AMD (FDR: 1.2×10^{-3}) and nvAMD (FDR: 3.33×10^{-6}). The second and third strongest associations were acute inflammatory response and immune response regulation (FDR: 2.9×10^{-2}) and antigen processing and presentation IFNy-mediated signaling pathways (FDR: 2×10^{-4}). Gene ontology for molecular functions revealed serine-type peptidase activity, cell adhesion molecule binding, and integrin binding in dry AMD (FDR $1.7–3.6 \times 10^{-3}$), while nvAMD showed enriched glycoprotein binding, receptor binding, and protein binding (FDR 8.9×10^{-3}). Using a cutoff of FC >2.0 and unadjusted P value < 0.05, they found 3 upregulated proteins (HEM2, RNAS1, and CPVL) and 26 downregulated proteins in dry AMD. Two of the 3 upregulated proteins (HEM2 and RNAS1) could not be validated by ELISA. In nvAMD, 3 proteins were downregulated (EPCR, BLMH, and APRV1) and 16 upregulated (top 3: VGFR1, 1433G, AL1A1). Two of the 16 were checked by ELISA (VGFR1 and CH3L1) but could not be validated. Gene set enrichment analysis analysis may identify potentially relevant gene networks, even if single protein network members do not meet stringent criteria for protein regulation. When the authors performed gene set enrichment analysis analysis, no pathway reached significance for dry AMD, but oxidative stress was the highest with an FDR of 0.261. All 4 oxidative stress–related proteins (GPX3, SOD1, SOD3, and CAT) showed upregulation trends. Gene set enrichment analysis analysis in nvAMD revealed 4 pathways with an FDR <0.05: focal adhesion, actin cytoskeleton regulation, myometrial relaxation and contraction, and NRF2 pathway.

Proteomics of RPE and Choroid in AMD Patients

Three Studies Used Either RPE Cells or RPE/Choroid Tissue from Cadaver Eyes. Karunadharma et al[27] analyzed the RPE proteome from cadaver eyes with no AMD (MGS 1) or with AMD (MGS 2–4) using 2D gel electrophoresis. Fifty healthy control donors (aged 29–91 years) were used to study the effect of aging on protein expression. They used 28 age-matched controls and compared them with 36 AMD (MGS 2, n = 11; MGS 3, n = 13; MGS 4, n = 12) patients to identify altered proteins. Twenty-four proteins changed with aging (4 of 24 increasing: cathepsin D, serotransferrin, prohibitin, and programmed cell death 6-interacting protein). In AMD, 23 proteins changed either at onset, linearly, or at the advanced stage. All except one (aldehyde dehydrogenase 9A1) increased with AMD stage. Seven proteins—ENO1, GLUL, HMGCS2, IDH1, PGAM1, RLBP1, and TUFM—exhibit distinct patterns in aging and AMD. These proteins decrease with age but increase in AMD at different stages.

Ingenuity pathway analysis revealed that while aging is associated with inhibition of glycolysis and gluconeogenesis, AMD is associated with the activation of both these pathways and the HIF1α pathway. Age-related macular degeneration is linked to inhibited oxidative phosphorylation, PPARα/RXRα activation, estrogen receptor signaling, and xenobiotic metabolism. Upstream regulator analysis indicates that the HSF1 stress response is suppressed across all AMD stages, while T cell receptor-mediated inflammation is activated in all stages.

Mcharg et al[28] examined submacular stromal tissue (RPE/choroid) from donors without AMD, grouped by homozygous high-risk (n = 8 each) or low-risk (n = 8) alleles at Chr1 and Chr10, plus 6 early AMD patients with high-risk Chr1 alleles only. In non-AMD donors, comparing Chr1 high-risk vs. low-risk identified 20 differentially expressed proteins, while Chr10 high-risk vs. low-risk found 17, with 7 overlapping, including top mast cell-related genes. Comparing Chr1 high-risk samples with and without AMD showed 56 of 769 proteins (7.3%) differentially expressed, with 43 upregulated and 13 downregulated in AMD. The top 5 altered GO biological processes were phospholipid efflux, high-density lipoprotein particle remodeling, plasma lipoprotein particle remodeling, reverse cholesterol transport, and plasma lipoprotein particle assembly. The top 5 proteins by FC (2–3.5×) were ITIH3, RNASE4, PLA2G2A, SFRP5, and APOC1. Other AMD-related proteins showed smaller changes, such as APOE (1.27× increase) and CFB (1.26× decrease). Proteins linked to other macular diseases also increased: C1QTNF5 (1.77×, late-onset retinal degeneration), EFEMP1 (1.29×, dominant drusen), and TIMP3 (1.33×, Sorsby fundus dystrophy). Neutrophil elastase and alpha-2-macroglobulin (macrophage-secreted) levels also rose. Ingenuity pathway analysis highlighted "IL-12 signaling and production in macrophages," "nitric oxide and reactive oxygen species production in macrophages," and "atherosclerosis signaling" as the top pathways in Chr1 high-risk AMD vs. Chr1 high-risk non-AMD patients.

Senabouth et al.[29] analyzed transcriptomics and proteomics of RPE cells from induced pluripotent stem cells of 44 GA patients and 39 healthy controls. Nine GA patients had ARMS2-HTRA1 haplotypes (absent in

Table 8. Most Frequently Encountered Proteins (All Biofluids)

Dry AMD			nvAMD		
Pathway	*Protein Superfamily*	*Proteins*	*Pathway*	*Protein Superfamily*	*Proteins*
Lipid metabolism and transport	Apolipoproteins	APOA1, APOA4, APOD	Lipoprotein metabolism and transport	Apolipoproteins	APOA1, APOA4, APOH
Protease regulation	Serpin protease inhibitors	SERPINA1, SERPINA4, SERPINA7, SERPINA10	Complement cascade Apoptosis	Complement system Chaperones	C3, C1RL, CFHR5, MASP2, CFI* Clusterin
Protease inhibitor	TIMPs	TIMP1, TIMP3*	Retinol transport, proteolysis	Lipocalin/retinol binding proteins	RBP, RPB3, RBP4, LCN1

AMD = age-related macular degeneration; nvAMD = neovascular age-related macular degeneration; TIMP = tissue inhibitor of metalloproteinases.
*Proteins included in the final validated proteomics presented by the authors.

controls), while 13 GA patients and 1 control had the CFH Y402H polymorphism (rs1061170). Retinal pigment epithelium induction yielded 5 cell populations; RPE6-matched adult RPE transcriptomics, and RPE4 was enriched in AMD-related transcripts. Proteomics revealed 234 differentially abundant proteins (FC $\geq$1.1, *t* test $P < 0.05$) between GA and control-induced pluripotent stem cell-RPE cells, with no protein exceeding 1.5× FC. The top 5 reduced proteins (up to 31%) were CRABP1, GAP43, CRYM, STMN2, and AKAP12, while the top upregulated (up to 47%) were CTAGE15, CD82, RHOF, PMEL, and TIMP3. The 10 most significant protein changes (FDR <0.05) were TIMP3, AMPH, EPS8L2, DR1, ARFGAP1, PPM1F, TRIO, CTTNBP2NL, RETSAT, and SH3GL1.

Tear Film

Winiarczyk et al conducted 2 studies[30,31] on tear film proteomes. In the first, they analyzed 8 nvAMD patients, 6 dry AMD patients, and 8 controls, using Schirmer tests to exclude dry eye syndrome. Tears were collected, and protein isolates run on 2D gel electrophoresis identified ~130 proteins per group. Eight proteins were exclusive to AMD, with 98 unique to dry AMD and 110 to nvAMD. In the second study, comparing 15 nvAMD patients and 15 controls, 469 protein spots were identified via 2D gel electrophoresis; 31 were differentially regulated, with 14 showing >1.5-FC, representing 11 proteins. Three (retinal dehydrogenase 1, adenosine triphosphate-dependent translocase ABCB1, and alpha-enolase) were upregulated (~1.5−2-fold), and 8 were downregulated.

Pathway Analysis of Significant Protein Changes

Differentially expressed proteins from each study, with |FC| >2 and statistical significance (Tables 5−7 and Supplementary Material), were analyzed using STRING[34] for protein−protein network meta-analysis and functional pathway analysis. Tables S1 to S3 (available at www.ophthalmologyscience.org) detail functional enrichment for selected proteins. Tables 5 and 6 present GO enrichment analysis for biological processes and molecular functions, respectively, highlighting top pathways. Gene ontology analysis leverages prior biological knowledge to link pathways to traits from genome-wide association study, reporting strength (Log10[observed/expected]) and signal (weighted harmonic mean of observed/expected ratio and -log[FDR]), adjusted with Benjamini−Hochberg FDR correction. Gene ontology analysis identified peptidase activity as a key recurring pathway in both nvAMD and dry AMD.

Kyoto Encyclopedia of Genes and Genomes[35] analysis (Table 7) mapped proteins to known pathways, identifying complement and coagulation cascades, cell adhesion molecules, fat digestion and absorption, and glycolysis/gluconeogenesis as AMD stage-specific pathways. Despite study heterogeneity, consistent protein changes appeared across multiple studies (Table 8). In dry AMD, apolipoproteins, serpin protease inhibitors, and TIMPs predominated across biofluids. In nvAMD, apolipoproteins, complement proteins,

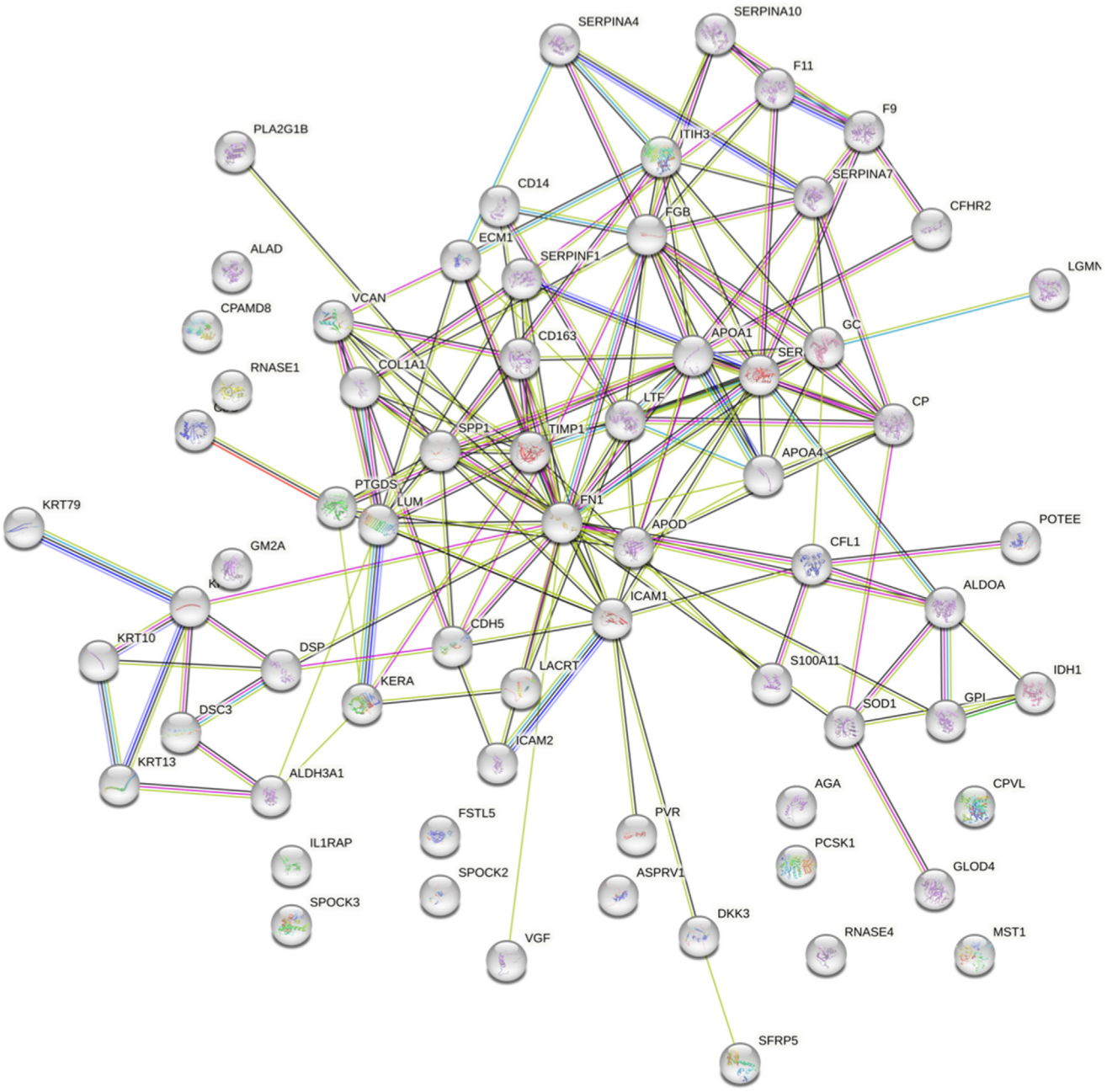


Figure 3. Protein–protein interaction (PPI) network in dry (e/i) AMD, FC $\geq|2|$. Protein–protein interaction network shows the associations among significantly altered proteins that exhibited a 2× FC between dry (e/i) AMD patients and controls, considering all types of samples (tear film studies did not provide any relevant proteins). Lipid metabolism (apolipoproteins) and protease regulation proteins (SERPINAs and TIMPs) are shown to have important functional links. PPI enrichment value of the network, $P < 1.0e\text{-}16$, means that the network has significantly more interactions than expected. The analyses include proteins from all biofluids (aqueous humor, vitreous, plasma, and urine) except tear film, as no proteins met the threshold ($|FC| \geq 2$) in this biofluid. AMD = age-related macular degeneration; e/i = early/intermediate; FC = fold change; TIMPs = tissue inhibitor of metalloproteinases.

chaperones (e.g., clusterin), retinol-binding protein, and lipocalin were most common. Complement proteins are more prevalent in nvAMD than dry AMD, indicating a significant role in its pathology, supported by increased exudation in complement inhibitor trials. Frequent detection of these proteins may reflect prioritization bias or reagent availability.

In late AMD (nvAMD with GA), unbiased analysis highlighted glycolysis/gluconeogenesis, posttranslational protein phosphorylation, and phospholipase inhibitor activity alongside complement pathways. Figures 3 to 6 illustrate protein–protein interaction networks and key pathways for dry and nvAMD. Two systematic reviews,[36,37] despite methodological differences, align with these findings,

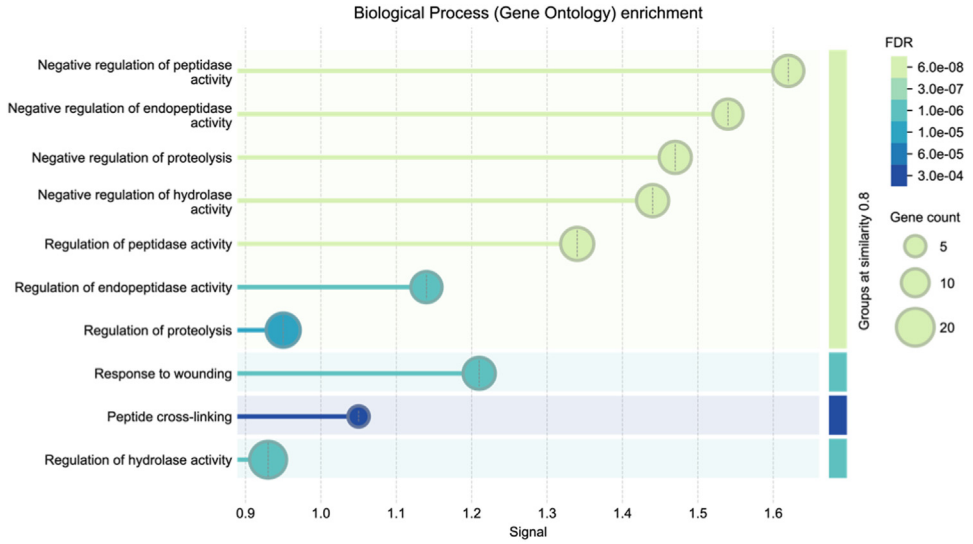


Figure 4. Gene ontology enrichment dry (e/i) AMD. Gene ontology analysis is based on previous PPIN analysis in dry (e/i) AMD, and the bubble plot shows the most significantly enriched pathways, x-axis: signal, which is defined as a weighted harmonic mean between the observed/expected ratio and -log(FDR). Y-axis: Pathways are grouped according to similarity, defined at 0.8. False discovery rate in color scale describes how significant the enrichment is, and *P* values are Benjamini–Hochberg adjusted. Bubble size corresponds to gene count for each enriched pathway. The analyses include proteins from all biofluids (aqueous humor, vitreous, plasma, and urine) except tear film, as no proteins met the threshold (|FC| ≥2) in this biofluid. AMD = age-related macular degeneration; e/i = early/intermediate; FC = fold change; FDR = false discovery rate; PPIN = protein–protein interaction network.

identifying lipid metabolism, complement, matrix remodeling, and proteins APOA1, C3, and LCN1 as significantly altered in AMD vs. controls to be an important and recurrent theme not only in nvAMD but also in dry AMD.

Discussion

Summary of Evidence

At the present review, 22 observational studies were included with a total of 6932 participants. The individual studies presented high clinical, methodological, and statistical heterogeneity. We tried to find out whether there are significant protein changes in different AMD types according to the type of sample and type of assay employed in proteomics.

Despite heterogeneity across studies, several proteins consistently showed significant changes and were validated as potential biomarkers for AMD (Table 8). These include apolipoproteins (APOA1, APOA4, and APOD in dry AMD; APOA1, APOA4, and APOH in nvAMD), complement-related proteins (CFHR5 and C3 in nvAMD), clusterin (in nvAMD), and protease inhibitors (SERPINA1, SERPINA4, and SERPINA7 in dry AMD). Apolipoprotein A1 (APOA1), a key component of high-density lipoprotein, regulates reverse cholesterol transport[38]—the process of removing cholesterol from tissues back to the liver. Its association with AMD highlights lipid metabolism dysregulation as a significant factor in the disease pathogenesis. Clusterin, an extracellular chaperone, modulates the complement system and contributes to processes like cell death, tumor progression, and neurodegenerative disorders.[39] It also participates in reverse cholesterol transport, with studies suggesting a role in lipid accumulation[40,41]—a hallmark of AMD-related drusen formation. However, these proteins, including apolipoproteins and clusterin, are also altered in other retinal diseases,[42–44] raising concerns about their specificity as AMD biomarkers. Serpins (serine protease inhibitors), such as SERPINA1, SERPINA4, and SERPINA7, regulate inflammation, coagulation, angiogenesis, and tissue homeostasis. Notably, SERPINA1 interacts with microglia and complement C3 in AMD, reducing oxidative stress and mediating immune responses.[45] The complement system, exemplified by C3 and CFHR5, is a well-established driver of AMD pathophysiology, yet linking specific complement fragments to therapeutic targets[46,47] remains challenging. These findings underscore the roles of lipid metabolism, complement activation, and protease regulation in AMD, though further research is needed to refine their biomarker and therapeutic potential.

Pathway analysis in AMD proteomics research reveals a complex network of biological processes driven by the diverse proteins identified in various biofluids. Despite this heterogeneity, STRING analysis has revealed specific pathways that are significantly enriched, pointing to their pivotal roles in AMD pathophysiology. Notably, lipid metabolism dysregulation stands out, with multiple lipid metabolism pathways consistently enriched across studies. This finding, supported by evidence from plasma samples in early/intermediate AMD (Table 9), strengthens the hypothesis that targeting lipid metabolism proteins or modulating cholesterol levels—either locally or systemically—could serve as a promising therapeutic

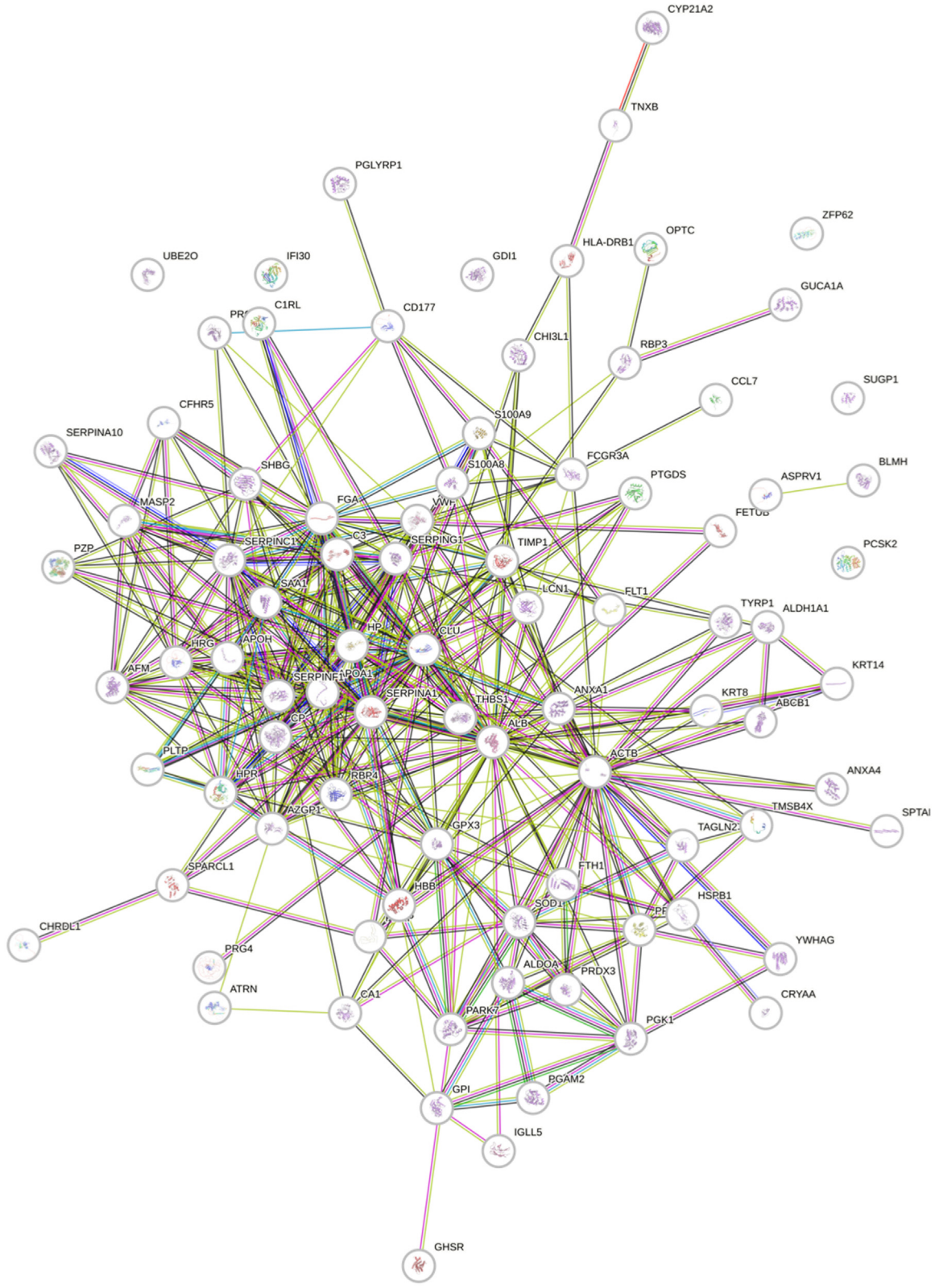


Figure 5. Protein–protein interaction network in nvAMD, FC ≥|2|. Protein–protein interaction network shows the associations among significantly altered proteins that exhibited a 2× FC between nvAMD patients and controls, considering all types of samples (RPE/choroidal tissue studies did not provide any relevant proteins). Lipoproteins, lipocalin, clusterin, and complement factors are shown to have several functional links. Protein–protein interaction enrichment value of the network, P < 1.0e-16, shows that the network has significantly more enrichments than expected. Analyses include proteins from all biofluids except RPE/choroid, which lacked proteins meeting the |FC| ≥2 criterion. FC = fold change; nvAMD = neovascular age-related macular degeneration; RPE = retinal pigment epithelium.

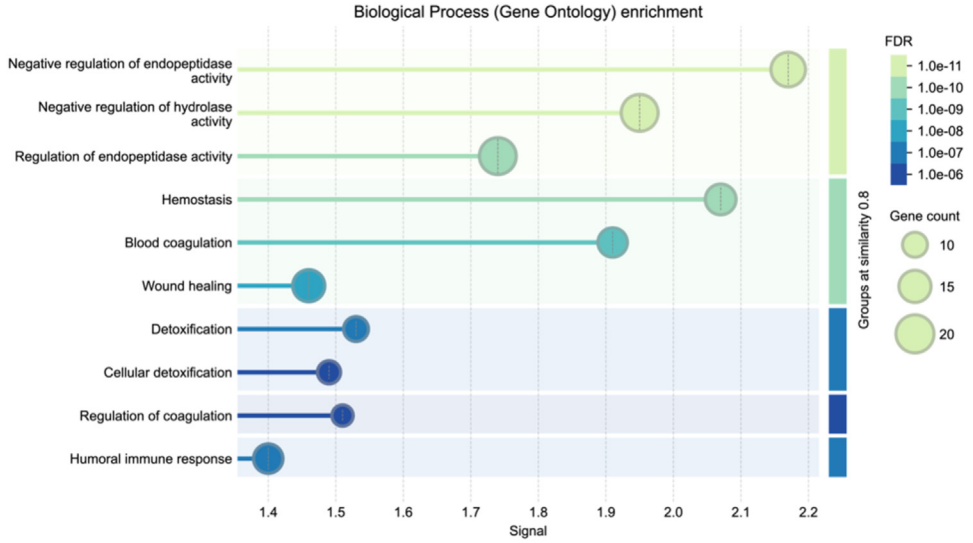


Figure 6. Gene ontology enrichment in nvAMD. Gene ontology analysis is based on previous PPIN analysis in nvAMD, and the bubble plot shows the most significantly enriched pathways in nvAMD, x-axis: Signal, y-axis: Groups at similarity. False discovery rate in color scale Benjamini–Hochberg adjusted. Bubble size corresponds to gene count for the enriched pathways. Analyses include proteins from all biofluids except RPE/choroid, which lacked proteins meeting the |FC| ≥2 criterion. FC = fold change; FDR = false discovery rate; nvAMD = neovascular age-related macular degeneration; PPIN = protein–protein interaction network; RPE = retinal pigment epithelium.

strategy for AMD. Beyond lipid metabolism, functional analyses also identified enriched pathways related to complement activation and endopeptidase activity, with AH and vitreous humor samples (especially in nvAMD) showing significant involvement of complement and coagulation cascades and peptidase regulation, respectively (Table 9). The fact that complement inhibitors increased exudation in all human clinical trials underscores the role of complement in nvAMD. Reinforcing the aforementioned pathway findings, Kyoto Encyclopedia of Genes and Genomes pathway analysis corroborated the results of GO analysis, thereby enhancing the confidence in the identified pathways' relevance to AMD. These observations underscore the multifaceted nature of AMD and suggest additional avenues for intervention. However, translating these insights into effective therapies requires larger interventional studies to pinpoint the precise conditions/stages and subgroups of AMD that these pathways should target.

Limitations

In general, selection bias was prevalent in most studies. This was primarily due to the small number of participants, which necessitated some studies to pool samples. Additionally, the proteins selected for validation were biased toward those that could be easily detected, those already reported in the literature to have significant biological implications, those commercially available with antibody kits, or, in some cases, the selection seemed random, not explained by the study methods section. As proteomic studies analyze hundreds to thousands of proteins, substantial sample sizes are required to have sufficient power to account for multiple testing. Only 1 study (Emilsson et al)[16] had the available sample size to perform appropriate Bonferroni-corrected statistics. It is also crucial to note that ocular biofluids were primarily studied in nvAMD patients, and there is a scarcity of studies involving ocular biofluids in dry AMD patients. Furthermore, most studies

Table 9. Pathways/Processes in Different AMD Stages (by Biofluid), Unbiased Proteomics FC >2

Sample Type	Total Number of Participants	e/i AMD	Late AMD - nvAMD (Pure GA Only 1 Study)
Aqueous humor	172	Endopeptidase inhibitor activity	Complement and coagulation cascades
Plasma	6019 (1 study 5457 alone)	Lipid metabolism	Glycolysis/glyconeogenesis Complement and coagulation cascades
Vitreous humor	254	Regulation of peptidase activity Cell adhesion	Regulation of protein activation cascade
RPE/choroid	233	Ns	-
Urine	84	Ns	Posttranslational protein phosphorylation
Tear film	52	Ns	Phospholipase inhibitor activity

AMD = age-related macular degeneration; e/i = early/intermediate; FC = fold change; GA = geographic atrophy; Ns = no significant enrichment detected; nvAMD = neovascular age-related macular degeneration; RPE = retinal pigment epithelium.

were cross-sectional, which limited their ability to capture the full course of disease progression and to distinguish between causality and mere association. Notably, Emilsson et al's[16] plasma study was the only one to examine causality by applying Mendelian randomization with available genetic data.

Other limitations inherent in each technique used also contribute to the challenges. For instance, MS-based techniques detect <5% of the total proteome and cannot detect low-abundance proteins. Aptamer-based techniques, on the other hand, can detect low-abundance proteins and almost a third of the total proteome; however, they lack absolute quantification. Additionally, there are inconsistencies in the cut-off criteria for protein identification and statistical analysis.

In this review, we conducted a comprehensive search for studies, focusing on more recent publications to provide the most up-to-date insights into the proteomics field in AMD. In recent years, proteomics discovery techniques have significantly improved, and the field has grown compared to previous periods. However, 4 studies[48–51] published before 2012 (the cutoff date used for this review) were not included, which could introduce a potential source of bias.

Overall, these limitations underscore the need for increased funding and collaboration to conduct appropriately designed studies with adequate sample sizes. This will enable a deeper understanding of the proteins and pathways involved in the different stages of AMD and its progression, potentially uncovering novel therapeutic targets.

Conclusion

Unbiased proteomics holds the promise of providing profound insights into the pathogenesis of AMD. However, the high cost and sensitivity of these techniques have limited their widespread application. Recent advancements, such as the use of aptamers, have enabled the identification of 5 to 10 times more proteins. Nevertheless, even this approach can only interrogate a third to a fifth of the entire proteome and cannot yet detect posttranslational modifications. Studies conducted to date have revealed significant heterogeneity in the individual results, reflecting the diverse nature of AMD. Functional analysis has demonstrated enrichment in specific pathways, such as lipid metabolism, protease regulation, and complement and coagulation cascades. We eagerly anticipate more extensive and well-powered studies involving ocular biofluids at all stages of AMD, which will pave the way for enhanced understanding and more effective therapeutics.

Footnotes and Disclosures

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Analysis and interpretation: Sideri, Correa, Ziakas, Tsinopoulos, Miller, Vavvas

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Overall responsibility: Sideri, Correa, Vavvas

Abbreviations and Acronyms:
2D = 2-dimensional; **AH** = aqueous humor; **AMD** = age-related macular degeneration; **AUC** = area under the curve; **CFH** = complement factor H; **CI** = confidence interval; **eAMD** = early age-related macular degeneration; **ELISA** = enzyme-linked immunosorbent assay; **FC** = fold change; **FDR** = false discovery rate; **GA** = geographic atrophy; **GO** = gene ontology; **LC** = liquid chromatography; **MGS** = Minnesota Grading System; **MS** = mass spectrometry; **nvAMD** = neovascular age-related macular degeneration; **RPE** = retinal pigment epithelium; **SDD** = subretinal drusenoid deposit; **TIMP** = tissue inhibitor of metalloproteinases.

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